Potential Drug-Drug Interactions in the Pediatric Intensive Care Unit

An Analysis using SafeICU and DDInter Drug Database

Project Report

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Abstract :

Potential Drug–Drug Interactions (DDIs) are a significant cause of adverse drug events in hospitalized patients, particularly among individuals receiving multiple medications. This study presents a patient-day level framework for identifying and analyzing potential DDIs using clinical data from the SafeICU database and the DDInter drug interaction database. Drug information was extracted from clinical notes, grouped by patient and prescription date, and all possible drug pairs were generated and matched against the DDInter database to identify confirmed interactions. The detected interactions were classified into Major, Moderate, and Minor severity levels, and a comprehensive patient-day dataset was constructed by integrating demographic information, number of prescribed drugs, and length of hospital stay.

Analysis of the constructed dataset included **6,941 patient-day prescription records** from **709 unique patients**. Among these, **435 patients (61.4%)** experienced at least one potential DDI, while **274 patients (38.6%)** had no identified interactions. The study population ranged in age from **0.17 to 18.85 years**, with a **median age of 5.36 years**. The most frequent drug used in the dataset was **Fentanyl . (piperacillin, vancomycin)** was the most frequently observed Major interaction, **[(dobutamine, vancomycin)]** was the most common Moderate interaction, and **[(dexamethasone, midazolam)]** was the most frequent Minor interaction. Overall, the developed framework provides an efficient and scalable approach for detecting, classifying, and analyzing potential DDIs, supporting improved medication safety and establishing a robust foundation for future predictive clinical decision-support systems.

Introduction

Drug–Drug Interactions (DDIs) occur when the effect of one medication is altered by the presence of another, potentially leading to reduced therapeutic efficacy, unexpected adverse drug reactions, or severe clinical complications. In Intensive Care Units (ICUs), patients are often prescribed multiple medications simultaneously to manage complex medical conditions, making them particularly susceptible to potential DDIs. Early identification of these interactions is essential for improving patient safety, reducing adverse events, and supporting informed clinical decision-making.

Pediatric ICU patients are especially vulnerable to DDIs because of age-related physiological differences, immature organ function, and the frequent use of multiple medications (polypharmacy). Even interactions considered moderate may have significant clinical consequences in critically ill children, making continuous monitoring of prescribed medications an important aspect of pediatric critical care.

The SafeICU database is a pediatric intensive care database developed under the SAFE-ICU Initiative through a collaboration involving AIIMS New Delhi and IIIT-Delhi, with support from the Indian Council of Medical Research (ICMR), the Centre of Excellence in Healthcare (COEH), and the Wellcome Trust/DBT India Alliance. It contains longitudinal clinical information, including patient demographics, admission details, laboratory measurements, vital signs, and clinical notes collected from critically ill pediatric patients. Access to the database is provided through a credentialing process to ensure patient privacy and responsible use of clinical data. The comprehensive nature of the dataset makes it a valuable resource for studying medication usage patterns, patient characteristics, and potential Drug–Drug Interactions (DDIs) in pediatric ICU settings.

The objective of this study is to develop a patient-day level framework for detecting and classifying potential Drug–Drug Interactions by integrating medication information extracted from SafeICU clinical notes with the DDInter interaction database. The study further analyzes the prevalence, severity distribution, commonly occurring interacting drug pairs, patient demographics, and clinical outcomes associated with potential DDIs.

Datasets Used

SafeICU Database -

This study utilized data from the **SafeICU** pediatric intensive care database, which contains longitudinal clinical records of critically ill children admitted to the Pediatric Intensive Care Unit (PICU). Three tables from the database were used to construct the patient-day Drug–Drug Interaction (DDI) dataset.

PATIENT Table -

The **PATIENT** table contains demographic information information such as **subject_id**, **gender**, **age_years**, and **expire_flag**, which was used to determine patient mortality. During data exploration, it was observed that the table contained duplicate records for some patients, which were removed during preprocessing.

ADMISSION Table -

The **ADMISSION** table stores admission-related information including **row_id , subject_id admitdate**, **admittime**, **dischdate**, **dischtime**, and **release_description**. Similar to the PATIENT table, duplicate admission records with identical admission and discharge details were identified and removed before further analysis to ensure one unique admission record per patient stay.

NOTE EVENTS Table -

The **NOTE EVENTS** table contains clinical notes, making it the largest table used in this study. Unlike the PATIENT and ADMISSION tables, it contained **substantially more records**, as each patient could have multiple clinical notes recorded on different days throughout their ICU stay. These notes included diagnoses, respiratory support information, and detailed treatment descriptions from which prescribed medications were extracted using text processing techniques. The extracted medications formed the basis for generating patient-day drug combinations and identifying potential DDIs.

DDInter Database -

The **DDInter** database was used as the reference source for identifying potential Drug–Drug Interactions. It is a comprehensive drug interaction knowledge base containing clinically validated interacting drug pairs along with their associated interaction severity and supporting evidence. For this study, the database was preprocessed by retaining the interacting drug names and the corresponding severity information (Major, Moderate, and Minor). After cleaning and standardization of drug names, the database was matched against drug pairs generated from the SafeICU clinical notes to identify confirmed potential DDIs.

Dataset Access -

Access to the SafeICU database was obtained through the official SafeICU data portal after completing the required credentialing process, including acceptance of the data use agreement. All analyses were conducted using de-identified patient data in accordance with the database access guidelines.

METHODOLOGY

1) Data Import and Preprocessing -

The SafeICU **PATIENT**, **ADMISSION**, and **NOTE EVENTS** tables, along with the **DDInter** database, were imported into the Kaggle Notebook environment using the Pandas library. Initial **data exploration** was performed to examine the **dataset dimensions**, **identify missing values**, **detect duplicate records**, and **verify data types** before further analysis.

Basic data cleaning was then carried out to improve data quality. In the **NOTE EVENTS** table, **records with missing chart_date values were removed** because the study was conducted at the **patient-day level**, and the **absence of a prescription date** made these records unsuitable for constructing daily medication histories and identifying drug interactions. Drug names extracted from the **clinical notes were standardized by converting them to lowercase to ensure consistent matching** with the DDInter database.

The **PATIENT** and **ADMISSION** tables were also examined for duplicate records. While multiple admissions for the same patient were expected, **some admission records contained identical subject_id, admitdate, and dischdate values, indicating exact duplicate entries.** These redundant records were removed to ensure that each hospitalization episode was represented only once. Similar duplicate records identified in the PATIENT table were also eliminated.

2) Drug Extraction and Construction of Patient-Day Medication Records -

Following data preprocessing, medication names were extracted from the free-text **description** column of the SafeICU **NOTE EVENTS** table using a **rule-based text extraction pipeline**. The pipeline first **converted all clinical notes to lowercase, removed punctuation and special characters, and tokenized the text into individual words**. These tokens were then **compared against a predefined reference list of valid drug names**, and **only exact matches were retained as medication mentions**. Duplicate drug names appearing within the same clinical note were removed to avoid repeated counting. This standardized extraction process identified a total of **82,259 drug mentions** across all clinical notes. The extracted medication records were subsequently grouped by **subject_id** and **chart_date**, enabling medication administration to be analyzed at the patient-day level.

To identify potential drug-drug interactions (DDIs), the DDInter database was first cleaned by retaining only the essential fields required for matching, including the names of the two interacting drugs, interaction severity, and supporting interaction information. A **lookup dictionary** was then constructed using **the drug pairs as keys and their corresponding interaction details as values**. This dictionary-based approach enabled rapid identification of interacting drug pairs while avoiding repeated searches through the complete DDInter database, thereby significantly improving computational efficiency.

3) Generation of Drug Combinations and DDI Matching -

The extracted medication records were subsequently transformed into **patient-day medication profiles**. First, all unique drugs administered to each patient on a given **chart_date** were aggregated into a single **drug_used** list . Next, **every possible two-drug combination was generated** from this list using the Python itertools.combinations() function, producing all potential drug pairs that could interact on that day . Finally, **each generated drug pair was compared against the previously constructed DDInter lookup** dictionary, and only those combinations present in the database were retained as **matched DDI pairs**. This process e**liminated non-interacting combinations and produced a curated set of clinically recognized potential drug-drug interactions for each patient-day**, forming the basis for subsequent severity classification and statistical analyses.

4) DDI Severity Classification and Integration with Admission Records

The matched drug pairs were subsequently classified according to the **severity information** provided in the DDInter database. For each patient-day, all matched interactions were grouped into three severity categories: **Major**, **Moderate**, and **Minor**. Separate columns (**ddi_major**, **ddi_moderate**, and **ddi_minor**) were created to store the corresponding interacting drug pairs, while additional columns (**num_major**, **num_moderate**, and **num_minor**) recorded the number of interactions in each severity category. This resulted in a comprehensive patient-day DDI profile containing both the interacting drug pairs and their severity-wise counts.

To incorporate hospitalization information, the patient-day DDI dataset was merged with the **ADMISSION** table. The merge was first performed using the common **subject_id**. Since a patient could have **multiple ICU admissions**, subject ID alone was insufficient to identify the correct hospitalization. Therefore, an admission record was retained only if the patient's **chart_date** satisfied the condition:

admitdate ≤ chart_date ≤ dischdate

This ensured that **every patient-day record was linked to its correct admission episode**. By applying this temporal matching criterion, each patient-day was assigned the appropriate admission details, including admission date, discharge date, and length of stay, while avoiding incorrect matches arising from multiple admissions for the same patient.

5) Calculation of Length of Stay and Construction of the Final Analysis Dataset -

Following successful integration of the admission records, the **Length of Stay (LOS)** for each hospitalization was calculated as the difference between the **discharge date** and the **admission date**, expressed in days. This variable was included to quantify the duration of each ICU stay and was subsequently used for descriptive and statistical analyses.

The resulting dataset was then merged with the **PATIENT** table to incorporate patient demographic information. Since a patient could have multiple admissions, the merge was performed using both **subject_id** and **row_id**, ensuring that each patient-day record was linked to the correct patient record. This step added demographic variables such as **age**, **gender**, and **mortality status (expire_flag)** while preserving the uniqueness of each admission episode.

After all preprocessing and integration steps, the final patient-day dataset was prepared for analysis. Each row represented a unique **patient-day**, and the dataset contained the following variables:

- **subject_id**
- **chart_date**
- **age**
- **gender**
- **admitdate**
- **admittime**
- **dischdate**
- **dischtime**
- **LOS_days (Length of Stay)**
- **drug_used** (list of all medications administered on that day)
- **num_drugs**
- **ddi_major**
- **ddi_moderate**
- **ddi_minor**
- **num_major**
- **num_moderate**
- **num_minor**
- **total_ddi**
- **mortality**

This finalized dataset served as the foundation for all subsequent analyses, including patient demographics, medication utilization, frequency of potential drug-drug interactions, severity-wise DDI analysis, length of stay comparisons, and mortality assessment.

6) Exploratory Data Analysis

Following the construction of the final patient-day dataset, an exploratory data analysis (EDA) was performed to summarize the characteristics of the study population and understand the distribution of potential drug–drug interactions. The analysis included:

- Identification of the total number of unique patients.
- Determination of the number of patients with and without at least one potential DDI.
- Analysis of the distribution of Major, Moderate, and Minor DDIs.

- Identification of the most frequently occurring drug–drug interaction pairs for each severity category.
- Analysis of the most commonly prescribed medications.
- Assessment of patient demographics, including age and gender distribution.

7) Multivariate Logistic Regression Analysis -

To identify factors independently associated with the occurrence of potential drug–drug interactions (DDIs), a multivariate logistic regression analysis was performed using the final patient-day dataset. Each row of the dataset, representing a unique patient-day prescription, was treated as an independent observation.

Prior to model development, only the variables required for the analysis **age, gender, number of prescribed drugs, mortality status, and DDI status** were retained. The dependent variable (DDI) was defined as a binary outcome, where **1** indicated the presence of at least one potential drug–drug interaction on a patient-day and **0** indicated the absence of any interaction. The dataset was examined for missing values, and observations containing missing predictor values were excluded to ensure complete-case analysis.

Gender was encoded as a binary categorical variable, whereas age and number of prescribed drugs were treated as continuous variables. An intercept term was added to the model, and a multivariate logistic regression model was fitted using the **Statsmodels** library in Python. Adjusted odds ratios (ORs) were obtained by exponentiating the regression coefficients, and corresponding **95% confidence intervals (CIs)** and **p-values** were calculated to assess the independent association between each predictor and the likelihood of experiencing a potential DDI. Statistical significance was determined using a two-sided **p-value < 0.05**.

RESULTS

Patient Characteristics -

The final analytical dataset consisted of **6941** patient-day prescriptions corresponding to **709** unique pediatric patients admitted to the intensive care unit. Patient demographic characteristics, including age and gender, were obtained from the PATIENT table after matching with the admission records. The study population included **60.22%** male patients and **39.77%** female patients. The age of patients ranged from **0.17** to **18.85** years, with a median age of **5.20** years.

Table 1 summarizes the demographic characteristics of the study population.

Table 1. Baseline demographic characteristics of the study population

Characteristic	Value
Total patient-day prescriptions	6941
Unique patients	709
Male	427
Female	282
Median age (years)	5.20
Minimum age (years)	0.17
Maximum age (years)	18.85

Prevalence of Potential Drug–Drug Interactions -

The occurrence of potential drug–drug interactions (DDIs) was assessed for each patient-day prescription in the constructed dataset. Patient-day records were classified as containing a potential DDI if at least one matched interaction was identified using the DDInter database. Overall, **435** patients experienced at least one potential DDI during their ICU stay, whereas **274** patients had no identified DDIs.

Potential DDIs were further categorized according to their severity into Major, Moderate, and Minor interactions based on the severity classification provided by the DDInter database. The distribution of DDIs across the three severity categories is presented in Table 2.

Table 2. Distribution of potential drug–drug interactions by severity

Severity Category	Number of DDIs	Percentage (%)
Major	1679	17.84
Moderate	6807	72.3
Minor	921	9.79
Total	9407	

Most Frequent Drug–Drug Interaction Pairs -

The frequency of identified potential DDIs was analyzed separately for each severity category. The most frequently occurring Major and Moderate interaction pairs are presented in Tables 3 and 4, respectively. These interaction pairs represent the combinations most commonly encountered in the study population and may warrant increased clinical attention during medication review.

Top 5 Most Frequent Major Drug Pairs

	Drug Pair	Frequency
0	(piperacillin, vancomycin)	298
1	(dexamethasone, fentanyl)	221
2	(fentanyl, fluconazole)	131
3	(fluconazole, midazolam)	112
4	(dexamethasone, ofloxacin)	86
5	(amikacin, furosemide)	85

Top 5 Most Frequent Moderate Drug Pairs

	Drug Pair	Frequency
0	(dobutamine, vancomycin)	315
1	(amikacin, pantoprazole)	238
2	(amikacin, piperacillin)	233
3	(amikacin, cefotaxime)	200
4	(furosemide, salbutamol)	180
5	(vancomycin, vecuronium)	176

Top 5 Most Frequent Minor Drug Pairs

	Drug Pair	Frequency
0	(dexamethasone, midazolam)	213
1	(budesonide, salbutamol)	156
2	(midazolam, prednisolone)	146
3	(methylprednisolone, midazolam)	71
4	(hydrocortisone, salbutamol)	51
5	(prednisolone, salbutamol)	33

Multivariate Logistic Regression

	Odds Ratio	CI Lower	CI Upper	P-value
const	0.033973	0.028245	0.040861	2.469007e-282
age	1.044013	1.029113	1.059128	4.284641e-09
gender	0.713931	0.631372	0.807284	7.688286e-08
num_drugs	1.816059	1.760748	1.873106	0.000000e+00
mortality	1.125942	0.991640	1.278433	6.718787e-02

Age

Age was significantly associated with the occurrence of potential drug–drug interactions (DDIs) (OR = **1.044**, 95% CI: **1.029–1.059**, $p < 0.001$). An odds ratio greater than 1 indicates a positive association, suggesting that with each one-year increase in age, the odds of experiencing a potential DDI increased by approximately **4.4%**. Since the 95% confidence interval does not include 1 and the p-value is well below 0.05, this association is statistically significant.

Inference: Older pediatric patients were at a slightly higher risk of experiencing potential DDIs.

Gender

Gender showed a statistically significant association with the occurrence of potential DDIs (OR = **0.714**, 95% CI: **0.631–0.807**, $p < 0.001$). An odds ratio less than 1 indicates that the category coded as **1** had approximately **28.6% lower odds** of experiencing a DDI compared with the reference category. The confidence interval does not include 1, confirming that this association is statistically significant.

Inference: Gender was independently associated with DDI occurrence. The exact interpretation (whether males or females had lower risk) depends on the coding used for the gender variable.

Number of Prescribed Drugs (Polypharmacy)

The number of prescribed drugs was the strongest predictor of potential DDIs (OR = **1.816**, 95% CI: **1.761–1.873**, $p < 0.001$). This indicates that each additional prescribed medication increased the odds of experiencing a potential DDI by approximately **81.6%**, after adjusting for age, gender, and mortality. The narrow confidence interval and highly significant p-value demonstrate a robust and precise association.

Inference: Polypharmacy was the most important independent risk factor for potential DDIs, indicating that patients receiving a larger number of medications were substantially more likely to experience drug interactions.

Mortality

Mortality was associated with an odds ratio of **1.126** (95% CI: **0.992–1.278**, $p = 0.067$). Although the estimated odds ratio suggests a **12.6% higher odds** of experiencing a potential DDI among patients who died, the **confidence interval includes 1** and the p-value is greater than the conventional significance threshold of 0.05.

Inference: Mortality was **not independently associated** with the occurrence of potential DDIs after adjusting for age, gender, and number of prescribed drugs.

Overall Interpretation

Multivariate logistic regression identified **age**, **gender**, and **number of prescribed drugs** as significant independent predictors of potential drug–drug interactions. Among these variables, **polypharmacy** demonstrated the strongest association with DDI occurrence. In contrast, **mortality did not show a statistically significant independent association** with potential DDIs after adjustment for the other covariates.

CONCLUSION

This study successfully identified potential drug–drug interactions (DDIs) in pediatric ICU patients by integrating clinical notes from the SafeICU dataset with the DDInter database. Polypharmacy emerged as the strongest independent predictor of DDIs, while age and gender also showed significant associations with DDI occurrence. These findings highlight the importance of routine DDI screening and careful medication review, particularly in patients receiving multiple drugs. Future work may incorporate dosage, treatment duration, and real-time clinical decision support systems to further improve medication safety in intensive care settings.